

1. 1번	2. 3번	3. 4번	4. 1번	5. 1번
6. 1번	7. 2번	8. 1번	9. 2번	10. 250
11. 1번	12. 3번	13. 4번	14. 2번	15. 3번
16. 4번	17. 1번	18. 4번	19. 4번	20. 4번
21. 3번	22. 2번	23. 2번	24. 4번	25. 3번
26. 1번	27. 3번	28. 5번	29. 4번	30. 2번
31. 1번	32. 3번	33. 4번	34. 4번	

$$1. U_f(P) = \frac{1}{2} \times 1 + \frac{1}{2} \times \frac{2}{3} + 2 \times \frac{1}{2} = \frac{11}{6}$$

$$2. n = 4 \text{ 이므로 } \Delta x = \frac{1}{2} \text{ 이다. 또한, } f(x) = \frac{1}{1+x} \text{ 이므로}$$

$$\begin{aligned} & \frac{\Delta x}{3} (1 \times y_0 + 4 \times y_1 + 2 \times y_2 + 4 \times y_3 + 1 \times y_4) \\ &= \frac{1}{3} \left(1 \times 1 + 4 \times \frac{2}{3} + 2 \times \frac{1}{2} + 4 \times \frac{2}{5} + 1 \times \frac{1}{3} \right) \\ &= \frac{11}{10} \text{ 이다.} \end{aligned}$$

3. 그래프를 그려보면 ①②③은 직선 $x = 2$ 에 대칭이므로

$$\int_{2-a}^2 f(x) dx = \int_2^{2+a} f(x) dx \text{ 를 만족한다.}$$

그러나 ④ $f(x) = (x-2)^3 - 1$ 은 직선 $x = 2$ 에 대칭이 아니고

따라서 $\int_{2-a}^2 f(x) dx = \int_2^{2+a} f(x) dx$ 을 만족하지 못한다.

4. $y = x - x^2$ 은 $x = 0$, $x = 1$ 에서 x 축과 만난다.
또한 그래프 개형을 그려보면 $0 \leq x \leq 1$ 일 때, S 는 최댓값을 갖는다.
즉, $a = 0$, $b = 1$ 이다.

5. 미적분학의 제 2 기본정리에 의하여

$$f\left(\frac{3}{2}\right) - f(0) = \int_0^{\frac{3}{2}} f'(x) dx = \frac{5}{8} \text{ 이다.}$$

$$\begin{aligned} 6. \int_0^2 |x^2 - x| dx &= - \int_0^1 x^2 - x dx + \int_1^2 x^2 - x dx \\ &= - \left[\frac{1}{3} x^3 - \frac{1}{2} x^2 \right]_0^1 + \left[\frac{1}{3} x^3 - \frac{1}{2} x^2 \right]_1^2 \\ &= - \left(\frac{1}{3} - \frac{1}{2} \right) + \left(\frac{8}{3} - 2 - \frac{1}{3} + \frac{1}{2} \right) \\ &= 1 \end{aligned}$$

$$7. \int_0^a \frac{\sqrt{a-x}}{\sqrt{x} + \sqrt{a-x}} dx = \frac{a}{2} \text{ 이므로}$$

$$\int_0^{2014} \frac{\sqrt{2014-x}}{\sqrt{x} + \sqrt{2014-x}} dx = 1007 \text{ 이다.}$$

$$8. \int_0^{\frac{\pi}{2}} \frac{\sin^n x}{\sin^n x + \cos^n x} dx = \frac{\pi}{4} \text{ 이므로}$$

$$\begin{aligned} & \int_0^{\frac{\pi}{2}} \sum_{k=1}^{100} \frac{\sin^k x}{\sin^k x + \cos^k x} dx \\ &= \int_0^{\frac{\pi}{2}} \frac{\sin x}{\sin x + \cos x} + \dots + \frac{\sin^{100} x}{\sin^{100} x + \cos^{100} x} dx \\ &= \frac{\pi}{4} + \frac{\pi}{4} + \dots + \frac{\pi}{4} = \frac{\pi}{4} \times 100 = 25\pi \text{ 이다.} \end{aligned}$$

$$9. \int_{-a}^a f(x) dx = \int_0^a f(x) + f(-x) dx \text{ 이므로}$$

$$\begin{aligned} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} f(x) dx &= \int_0^{\frac{\pi}{2}} f(x) + f(-x) dx \\ &= \int_0^{\frac{\pi}{2}} \cos x dx = 1 \text{ 이다.} \end{aligned}$$

$$\begin{aligned} 10. -100 \int_e^{e^2} f(x) dx \\ &= -100 \left[[x f(x)]_e^{e^2} - \int_e^{e^2} x f'(x) dx \right] (\because \text{부분적분}) \\ &= -100 \left\{ [e^2 f(e^2) - e f(e)] - \int_e^{e^2} \frac{\ln x}{x} dx \right\} \\ &= -100 \left(-1 - \int_1^{e^2} t dt \right) (\because \ln x = t \text{로 치환}) \\ &= -100 \left(-1 - \frac{1}{2} [t^2]_1^{e^2} \right) \\ &= 250 \end{aligned}$$

$$11. \int_a^b [f(x)]^2 dx = 1$$

$$\Leftrightarrow \left[x f^2(x) - \int x 2f(x) f'(x) dx \right]_a^b = 1 (\because \text{부분적분})$$

$$\Leftrightarrow b f^2(b) - a f^2(a) - \int_a^b x 2f(x) f'(x) dx = 1$$

$$\Leftrightarrow \int_a^b x f(x) f'(x) dx = -\frac{1}{2} (\because f(a) = f(b) = 0)$$

$$12. \int_0^2 g(2-x) dx \text{ 에서 } 2-x = t \text{로 치환하면}$$

$$\int_0^2 g(2-x) dx = \int_0^2 g(t) dt \text{ 이다.}$$

$$\text{또한, } 4 \int_0^1 g(2x) dx \text{ 에서 } 2x = \theta \text{로 치환하면}$$

$$4 \int_0^1 g(2x) dx = 2 \int_0^2 g(\theta) d\theta \text{ 이다.}$$

$$\text{그리고 } \int_0^4 g(x) dx = -3, \int_2^4 g(x) dx = \frac{2}{3} \text{ 이므로}$$

$$\int_0^2 g(x) dx = \int_0^4 g(x) dx - \int_2^4 g(x) dx = -\frac{11}{3} \text{ 이다.}$$

$$\begin{aligned} \text{즉, } \int_0^2 g(2-x) dx + 4 \int_0^1 g(2x) dx \\ &= \int_0^2 g(t) dt + 2 \int_0^2 g(\theta) d\theta = 3 \int_0^2 g(x) dx = -11 \text{ 이다.} \end{aligned}$$

13. $\int_a^b f(x)dx + \int_{f(a)}^{f(b)} f^{-1}(x)dx = bf(b) - af(a)$ 이므로
 $\int_1^3 f(x)dx + \int_2^5 f^{-1}(x)dx = 13$ 이고
 $\Rightarrow \int_2^5 f^{-1}(x)dx = 13 - \int_1^3 f(x)dx = \frac{17}{2}$ 이다.

[다른 풀이]

$$\begin{aligned} \int_2^5 f^{-1}(y)dy &= \int_1^3 xf'(x)dx \quad (\because f^{-1}(y) = x) \\ &= [xf(x)]_1^3 - \int_1^3 f(x)dx \quad (\because \text{부분적분}) \\ &= 3f(3) - f(1) - \frac{9}{2} = \frac{17}{2} \text{ 이다.} \end{aligned}$$

14. $\int_a^b f(x)dx + \int_{f(a)}^{f(b)} f^{-1}(x)dx = bf(b) - af(a)$ 이므로
 $\int_0^1 f(x)dx + \int_0^2 f^{-1}(x)dx = 2$ 이고
 $\Rightarrow \int_0^2 f^{-1}(x)dx = 2 - \int_0^1 f(x)dx$
 $= 2 - \int_0^1 x^3 + x dx = 2 - \left[\frac{1}{4}x^4 + \frac{1}{2}x^2 \right]_0^1 = \frac{5}{4}$ 이다.

[다른 풀이]

$$\begin{aligned} f(0) &= 0, \quad f(1) = 2 \text{ 이므로} \\ \int_0^2 f^{-1}(x)dx &= \int_0^1 tf'(t)dt \quad (\because f^{-1}(x) = t) \\ &= \int_0^1 3t^3 + t dt = \left[\frac{3}{4}t^4 + \frac{1}{2}t^2 \right]_0^1 = \frac{5}{4} \text{ 이다.} \end{aligned}$$

15. $\lim_{n \rightarrow \infty} \frac{2}{n} \sum_{k=1}^n \left[\left(1 + \frac{2k}{n}\right)^2 - \left(1 + \frac{2k}{n}\right) \right]$
 $= \int_0^2 (1+x)^2 - (1+x)dx \quad \left(\because \frac{2k}{n} = x, \frac{2}{n} = dx \right)$
 $= \left[\frac{1}{3}(1+x)^3 - \frac{1}{2}(1+x)^2 \right]_0^2 = \frac{14}{3}$

16. $\lim_{n \rightarrow \infty} \left(\frac{1}{\sqrt{n}\sqrt{n+1}} + \frac{1}{\sqrt{n}\sqrt{n+2}} + \dots + \frac{1}{\sqrt{n}\sqrt{n+n}} \right)$
 $= \lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{1}{\sqrt{n}\sqrt{n+k}} = \lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{1}{\sqrt{1+\frac{k}{n}}} \times \frac{1}{n}$
 $= \int_0^1 \frac{1}{\sqrt{1+x}} dx \quad \left(\because \frac{k}{n} = x, \frac{1}{n} = dx \right)$
 $= 2 \left[\sqrt{1+x} \right]_0^1 = 2\sqrt{2} - 2$

17. $\lim_{n \rightarrow \infty} \frac{(1+2^2+3^2+\dots+n^2)^2}{n(1+2^4+3^4+\dots+n^4)}$
 $= \lim_{n \rightarrow \infty} \frac{\sum_{k=1}^n k^2 \sum_{k=1}^n k^2}{n \sum_{k=1}^n k^4} = \lim_{n \rightarrow \infty} \frac{\sum_{k=1}^n \left(\frac{k}{n}\right)^2 \frac{1}{n} \sum_{k=1}^n \left(\frac{k}{n}\right)^2 \frac{1}{n}}{\sum_{k=1}^n \left(\frac{k}{n}\right)^4 \frac{1}{n}}$
 $= \frac{\int_0^1 x^2 dx \int_0^1 x^2 dx}{\int_0^1 x^4 dx} \quad \left(\because \frac{k}{n} = x, \frac{1}{n} = dx \right)$
 $= \frac{\frac{1}{3}[x^3]_0^1 \times \frac{1}{3}[x^3]_0^1}{\frac{1}{5}[x^5]_0^1} = \frac{5}{9}$

[다른 풀이]

n 제곱의 합 공식을 이용하면

$$\lim_{n \rightarrow \infty} \frac{(1+2^2+3^2+\dots+n^2)^2}{n(1+2^4+3^4+\dots+n^4)} = \lim_{n \rightarrow \infty} \frac{\left(\frac{1}{3}n^3 + \dots\right)^2}{n\left(\frac{1}{5}n^5 + \dots\right)} = \frac{5}{9} \text{ 이다.}$$

18. $\lim_{n \rightarrow \infty} \frac{7}{\sqrt[3]{n^2}} \sum_{k=1}^n \frac{1}{\sqrt[3]{n+7k}}$
 $= \lim_{n \rightarrow \infty} \frac{7}{n} \sum_{k=1}^n \frac{1}{\sqrt[3]{1+\frac{7k}{n}}} \quad \left(\because \frac{7k}{n} = x, \frac{7}{n} = dx \right)$
 $= \int_0^7 \frac{1}{(1+x)^{\frac{1}{3}}} dx = \int_0^7 (1+x)^{-\frac{1}{3}} dx = \frac{3}{2} \left[(1+x)^{\frac{2}{3}} \right]_0^7 = \frac{9}{2}$

19. $\lim_{n \rightarrow \infty} \frac{1}{n} \left(\sqrt{3} + \sqrt{3+\frac{1}{n}} + \sqrt{3+\frac{2}{n}} + \dots + \sqrt{3+\frac{n-1}{n}} \right)$
 $= \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=0}^{n-1} \sqrt{3+\frac{k}{n}} \quad \left(\because \frac{k}{n} = x, \frac{1}{n} = dx \right)$
 $= \int_0^1 \sqrt{3+x} dx$
 $= \frac{2}{3} \left[(3+x)^{\frac{3}{2}} \right]_0^1 = \frac{2}{3} (8 - 3\sqrt{3})$

20. $\lim_{n \rightarrow \infty} \frac{1}{n} \left(\cos \frac{2}{3n} + \cos \frac{4}{3n} + \cos \frac{6}{3n} + \dots + \cos \frac{6n}{3n} \right)$
 $= \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^{3n} \cos \frac{2k}{3n} \quad \left(\because \frac{2k}{3n} = x, \frac{2}{3n} = dx \right)$
 $= \frac{3}{2} \int_0^2 \cos(x) dx = \frac{3}{2} [\sin(x)]_0^2 = \frac{3}{2} \sin 2$

21. $\lim_{n \rightarrow \infty} \frac{K_{2n+2}}{K_{2n}} = \lim_{n \rightarrow \infty} \frac{\int_0^{\frac{\pi}{2}} \sin^{2n+2} x dx}{\int_0^{\frac{\pi}{2}} \sin^{2n} x dx}$
 $= \lim_{n \rightarrow \infty} \frac{\frac{2n+1}{2n+2} \times \frac{2n-1}{2n} \times \dots \times \frac{1}{2} \times \frac{\pi}{2}}{\frac{2n-1}{2n} \times \frac{2n-3}{2n-2} \times \dots \times \frac{1}{2} \times \frac{\pi}{2}} \quad (\because \text{왈리스 정리})$
 $= \lim_{n \rightarrow \infty} \frac{2n+1}{2n+2} = 1$

22. $\frac{\int_1^3 x^2 - 4x + 3 dx}{3-1} = \frac{\left[\frac{1}{3}x^3 - 2x^2 + 3x \right]_1^3}{2} = -\frac{2}{3}$ 이므로
 평균값이는 $\frac{2}{3}$ 이다.

23. $f(x) = \frac{1}{1+x^3}$ 은 구간 $[0,1]$ 에서 연속이므로

적분의 평균값 정리를 이용하면 다음이 성립한다.

$$\frac{\int_a^b \frac{1}{1+x^3} dx}{b-a} = \frac{1}{1+c^3}, \quad 0 \leq a < c \leq b < 1$$

또한, $0 < c < 1$ 이므로

$$\Rightarrow 0 < c^3 < 1 \Rightarrow 1 < 1+c^3 < 2 \Rightarrow 1 > \frac{1}{1+c^3} > \frac{1}{2} \text{ 이다.}$$

따라서 $\frac{1}{2} < \frac{\int_a^b \frac{1}{1+x^3} dx}{b-a} < 1$ 이다.

24. $F(x) = \int_0^{x^2} t \cos t^2 dt$ 에서 정적분과 도함수를 이용하면
 $F'(x) = x^2 \cos(x^4) \times 2x = 2x^3 \cos(x^4)$ 이다.

$$\begin{aligned} 25. f(x) &= \int_{-\sin^{-1}\sqrt{x}}^{\sin^{-1}\sqrt{x}} \sin \sqrt{|t|} dt \\ &= 2 \int_0^{\sin^{-1}\sqrt{x}} \sin \sqrt{|t|} dt \quad (\because \sin \sqrt{|t|} \text{는 우함수}) \\ &= 2 \int_0^{\sin^{-1}\sqrt{x}} \sin \sqrt{t} dt \quad (\because t \geq 0) \end{aligned}$$

여기서 정적분과 도함수를 이용하면

$$f'(x) = 2 \sin \sqrt{\sin^{-1}\sqrt{x}} \times \frac{1}{\sqrt{1-x}} \times \frac{1}{2\sqrt{x}} \text{이다.}$$

$$\therefore f'\left(\frac{1}{2}\right) = 2 \sin \frac{\sqrt{\pi}}{2}$$

$$26. H(x) = \frac{1}{x} \int_3^x [2t - 3H'(t)] dt$$

$$H'(x) = -\frac{1}{x^2} \int_3^x [2t - 3H'(t)] dt + \frac{1}{x} [2x - 3H'(x)] \text{이다.}$$

$$\text{따라서 } H'(3) = 2 - H'(3) \Rightarrow H'(3) = 1 \text{이다.}$$

27. 적분식 안에 미분 변수 x 가 있으므로 라이프니츠 정리를 이용하자.

$$f(x) = \int_0^x (e^x - t) e^t dt$$

$$f'(x) = (e^x - x) e^x + \int_0^x e^x e^t dt$$

$$\therefore f'(0) = 1$$

[다른 풀이]

$$f(x) = \int_0^x (e^x - t) e^t dt = e^x \int_0^x e^t dt - \int_0^x t e^t dt \text{이므로}$$

양변을 x 에 대해 미분하면

$$f'(x) = e^x \int_0^x e^t dt + e^x \cdot e^x - x e^x \text{이다.}$$

$$\text{즉, } f'(0) = 1 \text{이다.}$$

$$28. f(x) = \int_{2x}^{3x} \frac{1}{\sin(t-x)} dt \quad (\because t-x = \theta)$$

$$= \int_x^{2x} \frac{1}{\sin(\theta)} d\theta$$

$$f'(x) = \frac{1}{\sin 2x} \times 2 - \frac{1}{\sin x}$$

$$\therefore f'\left(\frac{\pi}{6}\right) = \frac{4}{\sqrt{3}} - 2$$

$$29. \lim_{x \rightarrow 0^+} \frac{\int_x^{3x} \frac{\sin 3t}{t} dt}{x}$$

$$= \lim_{x \rightarrow 0^+} \frac{\frac{\sin 9x}{3x} \times 3 - \frac{\sin 3x}{x} \times 1}{1} \quad (\because \text{로피탈 정리})$$

$$= \lim_{x \rightarrow 0^+} \frac{\sin 9x}{9x} \times 9 - \frac{\sin 3x}{3x} \times 3$$

$$= 6$$

[다른 풀이]

$$f(t) = \frac{\sin 3t}{t} \text{는 구간 } [x, 3x] \text{에서 연속이므로}$$

적분의 평균값 정리를 이용하면 다음이 성립한다.

$$\frac{\int_x^{3x} \frac{\sin 3t}{t} dt}{2x} = \frac{\sin 3c}{c}, \quad x < c < 3x$$

$$\text{이때, } x \rightarrow 0^+ \text{이면 } c \rightarrow 0^+ \text{이다.}$$

$$\begin{aligned} \text{즉, } \lim_{x \rightarrow 0^+} \frac{\int_x^{3x} \frac{\sin 3t}{t} dt}{x} &= \lim_{x \rightarrow 0^+} \frac{2 \int_x^{3x} \frac{\sin 3t}{t} dt}{2x} \\ &= \lim_{x \rightarrow 0^+} 2 \frac{\sin 3c}{c} = 2 \cdot 3 = 6 \text{이다.} \end{aligned}$$

$$30. \lim_{x \rightarrow \frac{\pi}{2}} \frac{x \int_{\frac{\pi}{2}}^x \frac{\sin t}{t} dt}{4x^2 - \pi^2}$$

$$= \lim_{x \rightarrow \frac{\pi}{2}} \frac{\int_{\frac{\pi}{2}}^x \frac{\sin t}{t} dt + \sin x}{8x} \quad (\because \text{로피탈 정리})$$

$$= \frac{1}{4\pi}$$

[다른 풀이]

$$f(t) = \frac{\sin t}{t} \text{는 구간 } \left[\frac{\pi}{2}, x\right] \text{에서 연속이므로}$$

적분의 평균값 정리를 이용하면 다음이 성립한다.

$$\frac{\int_{\frac{\pi}{2}}^x \frac{\sin t}{t} dt}{x - \frac{\pi}{2}} = \frac{\sin c}{c}, \quad \frac{\pi}{2} < c < x$$

$$\text{이때, } x \rightarrow \frac{\pi}{2} \text{이면 } c \rightarrow \frac{\pi}{2} \text{이다.}$$

$$\begin{aligned} \text{즉, } \lim_{x \rightarrow \frac{\pi}{2}} \frac{x \int_{\frac{\pi}{2}}^x \frac{\sin t}{t} dt}{4x^2 - \pi^2} &= \lim_{x \rightarrow \frac{\pi}{2}} \frac{x \int_{\frac{\pi}{2}}^x \frac{\sin t}{t} dt}{(2x - \pi)(2x + \pi)} \\ &= \lim_{x \rightarrow \frac{\pi}{2}} \frac{x \int_{\frac{\pi}{2}}^x \frac{\sin t}{t} dt}{2(x - \frac{\pi}{2})(2x + \pi)} = \lim_{x \rightarrow \frac{\pi}{2}} \frac{x \frac{\sin c}{c}}{2(2x + \pi)} = \frac{\frac{\pi}{2} \cdot \frac{2}{\pi}}{2(2 \cdot \frac{\pi}{2} + \pi)} = 4\pi \text{이다.} \end{aligned}$$

$$31. f(t) = \frac{1}{2} (\sinh t \cosh t - 1) - \int_1^{\cosh t} \sqrt{x^2 - 1} dx$$

$$\begin{aligned} f'(t) &= \frac{1}{2} (\cosh^2 t + \sinh^2 t) - \sqrt{\cosh^2 t - 1} \times \sinh t \\ &= \frac{1}{2} (\cosh^2 t - \sinh^2 t) = \frac{1}{2} \end{aligned}$$

$$\text{따라서 } f'(t) = \frac{1}{2} \text{이고 적분하면 } f(t) = \frac{1}{2}t + C \text{이다.}$$

$$\text{또한 } f(0) = -\frac{1}{2} \text{이므로 } f(t) = \frac{1}{2}t - \frac{1}{2} \text{이고 } f(2) = \frac{1}{2} \text{이다.}$$

32. 극대점을 찾기 위해서 임계점을 구하자.

$$f(x) = \int_0^x \frac{\sin t}{t} dt$$

$$f'(x) = \frac{\sin x}{x} = 0 \quad \therefore x = \pi, 2\pi, 3\pi, \dots$$

$$f''(x) = \frac{\cos x \times x - \sin x}{x^2}$$

$$\text{즉, } x = (2n-1)\pi \text{이면 } f''(x) < 0 \text{이므로 극대값을 갖는다.}$$

$$33. 6 + \int_a^x \frac{f(t)}{t^2} dt = 2\sqrt{x}$$

$$1) \text{ 양변에 } x = a \text{를 대입하면 } 6 = 2\sqrt{a} \text{이다. 즉, } a = 9$$

$$2) \text{ 양변을 미분하면 } \frac{f(x)}{x^2} = \frac{1}{\sqrt{x}} \Rightarrow f(x) = x^{\frac{3}{2}} \text{이다.}$$

$$\text{따라서 } f(a) = a^{\frac{3}{2}} = 3^3 \text{이다.}$$

34. $f(x) = \cos x + \int_0^{\frac{\pi}{2}} f(x) dx$ 에서 $\int_0^{\frac{\pi}{2}} f(x) dx$ 는 상수값이므로

$$\int_0^{\frac{\pi}{2}} f(x) dx = c \text{ 라 하면}$$

$$f(x) = \cos x + c \text{ 이다.}$$

$$\int_0^{\frac{\pi}{2}} f(x) dx = c$$

$$\Leftrightarrow \int_0^{\frac{\pi}{2}} (\cos x + c) dx = c$$

$$\Leftrightarrow [\sin x + cx]_0^{\frac{\pi}{2}} = c$$

$$\Leftrightarrow 1 + \frac{\pi}{2}c = c \quad \therefore c = \frac{2}{2-\pi}$$

$$\text{따라서 } f(x) = \cos x + \frac{2}{2-\pi} \text{ 이고 } f(0) = \frac{4-\pi}{2-\pi} \text{ 이다.}$$



장황수학

1. 3번	2. 1번	3. 2번	4. 3번	5. 4번
6. 1번	7. 2번	8. 1번	9. 4번	10. 2번
11. 4번	12. 4번	13. 4번	14. 1번	15. 2번
16. 4번	17. 4번	18. 3번		

1. $a_{n+2} = 2a_{n+1} + a_n$ 은 $\frac{a_{n+2}}{a_{n+1}} = 2 + \frac{a_n}{a_{n+1}}$ 이므로

$$\lim_{n \rightarrow \infty} b_n = \lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} = \lim_{n \rightarrow \infty} \frac{a_{n+2}}{a_{n+1}} = \alpha \text{ 라 하면}$$

$$\lim_{n \rightarrow \infty} \frac{a_{n+2}}{a_{n+1}} = \lim_{n \rightarrow \infty} 2 + \frac{a_n}{a_{n+1}}$$

$$\Rightarrow \alpha = 2 + \frac{1}{\alpha}$$

$$\Rightarrow \alpha^2 - 2\alpha - 1 = 0$$

$$\Rightarrow \alpha = 1 + \sqrt{2} \text{ 또는 } 1 - \sqrt{2}$$

$$\Rightarrow a_3 = 2a_2 + a_1 = 3,$$

$$a_4 = 2a_3 + a_2 = 7,$$

$$a_5 = 2a_4 + a_3 = 17,$$

⋮

따라서 $\frac{a_4}{a_3} = \frac{7}{3}, \frac{a_5}{a_4} = \frac{17}{7}, \dots$ 이므로 $\lim_{n \rightarrow \infty} b_n$ 의 값은 $1 + \sqrt{2}$ 이다.

2. $\sum_{n=1}^{\infty} \frac{1}{n(n+1)} = \lim_{k \rightarrow \infty} \sum_{n=1}^k \frac{1}{n(n+1)} = \lim_{k \rightarrow \infty} \sum_{n=1}^k \left(\frac{1}{n} - \frac{1}{n+1} \right)$
 $= \lim_{k \rightarrow \infty} \left(1 - \frac{1}{2} + \left(\frac{1}{2} - \frac{1}{3} \right) + \left(\frac{1}{3} - \frac{1}{4} \right) + \dots + \left(\frac{1}{k} - \frac{1}{k+1} \right) \right)$
 $= \lim_{k \rightarrow \infty} 1 - \frac{1}{k+1} = 1$

3. $\sum_{n=2}^{\infty} \ln \frac{(n-1)(n+1)}{n^2} = \lim_{k \rightarrow \infty} \sum_{n=2}^k \ln \frac{(n-1)(n+1)}{n^2}$
 $= \lim_{k \rightarrow \infty} \sum_{n=2}^k \ln \frac{n-1}{n} + \ln \frac{n+1}{n}$
 $= \lim_{k \rightarrow \infty} \left(\ln \frac{1}{2} + \ln \frac{3}{2} + \left(\ln \frac{2}{3} + \ln \frac{4}{3} \right) + \left(\ln \frac{3}{4} + \ln \frac{5}{4} \right) \right.$
 $\left. + \dots + \left(\ln \frac{k-1}{k} + \ln \frac{k+1}{k} \right) \right)$
 $= \lim_{k \rightarrow \infty} \ln \frac{1}{2} + \ln \frac{k+1}{k} = \ln \frac{1}{2}$

4. $\lim_{n \rightarrow \infty} \left(\frac{8}{n} + \frac{8n+1}{n^2+1} + \frac{8n+2}{n^2+4} + \frac{8n+3}{n^2+9} + \dots + \frac{9n-1}{2n^2-2n+1} \right)$
 $= \lim_{n \rightarrow \infty} \sum_{k=0}^{n-1} \frac{8n+k}{n^2+k^2} = \lim_{n \rightarrow \infty} \sum_{k=0}^{n-1} \frac{\frac{8n+k}{n^2}}{\frac{n^2+k^2}{n^2}} = \lim_{n \rightarrow \infty} \sum_{k=0}^{n-1} \frac{8 + \frac{k}{n}}{1 + \left(\frac{k}{n} \right)^2} \cdot \frac{1}{n}$
 $= \int_0^1 \frac{8+x}{1+x^2} dx = \left[8 \tan^{-1} x + \frac{1}{2} \ln(1+x^2) \right]_0^1 = 2\pi + \frac{1}{2} \ln 2$

5. $\lim_{n \rightarrow \infty} \sum_{k=n+1}^{2n} \frac{1}{\sqrt{4n^2-k^2}} = \lim_{n \rightarrow \infty} \sum_{k=n+1}^{2n} \frac{1}{\sqrt{4n^2-k^2}} \cdot \frac{1}{n}$
 $= \lim_{n \rightarrow \infty} \sum_{k=n+1}^{2n} \frac{1}{\sqrt{4 - \left(\frac{k}{n} \right)^2}} \cdot \frac{1}{n} = \int_1^2 \frac{1}{\sqrt{4-x^2}} dx \left(\because \frac{k}{n} = x \right)$
 $= \left[\sin^{-1} \frac{x}{2} \right]_1^2 = \frac{\pi}{2} - \frac{\pi}{6} = \frac{\pi}{3}$

6. $\lim_{n \rightarrow \infty} \frac{\sqrt[n]{(n+1)(n+2) \cdots (2n-1)(2n)}}{n}$
 $= \lim_{n \rightarrow \infty} \left(\frac{(n+1)(n+2) \cdots (2n-1)(2n)}{n^n} \right)^{\frac{1}{n}}$
 $= \lim_{n \rightarrow \infty} e^{\frac{1}{n} \ln \left(\frac{(n+1)(n+2) \cdots (2n-1)(2n)}{n^n} \right)^{\frac{1}{n}}}$
 $= \lim_{n \rightarrow \infty} e^{\frac{1}{n} \ln \left(\frac{(n+1)(n+2) \cdots (2n-1)(2n)}{n^n} \right)}$
 $= \lim_{n \rightarrow \infty} e^{\frac{1}{n} \left(\ln \frac{n+1}{n} + \ln \frac{n+2}{n} + \ln \frac{n+3}{n} + \dots + \ln \frac{n+n}{n} \right)}$
 $= \lim_{n \rightarrow \infty} e^{\frac{1}{n} \sum_{k=1}^n \ln \frac{n+k}{n}} = \lim_{n \rightarrow \infty} e^{\frac{1}{n} \sum_{k=1}^n \ln \left(1 + \frac{k}{n} \right)} = e^{\int_0^1 \ln(1+x) dx}$
 $= e^{[(1+x) \ln(1+x) - (1+x)]_0^1} = e^{2 \ln 2 - 1} = \frac{4}{e}$

7. $\lim_{x \rightarrow 0} \frac{\int_0^{2x^2} \sin \left(\frac{t^2}{3} \right) dt}{x^6} = \lim_{x \rightarrow 0} \frac{\sin \left(\frac{4x^4}{3} \right) \times 4x}{6x^5} = \lim_{x \rightarrow 0} \frac{\sin \left(\frac{4x^4}{3} \right) \times 2}{3x^4}$
 $= \lim_{x \rightarrow 0} \frac{\left(\frac{4x^4}{3} + \dots \right) \times 2}{3x^4} = \frac{8}{9}$

[다른 풀이]

$$\lim_{x \rightarrow 0} \frac{\int_0^{2x^2} \sin \left(\frac{t^2}{3} \right) dt}{x^6} = \lim_{x \rightarrow 0} \frac{\int_0^{2x^2} \frac{t^2}{3} - \frac{1}{3!} \left(\frac{t^2}{3} \right)^3 + \dots dt}{x^6}$$

$$= \lim_{x \rightarrow 0} \frac{\left[\frac{t^3}{9} - \dots \right]_0^{2x^2}}{x^6} = \lim_{x \rightarrow 0} \frac{\frac{8x^6}{9} - \dots}{x^6} = \frac{8}{9}$$

[참고]

해당 문제는 적분의 평균값정리로 풀 수 없다.

$f(t) = \sin \left(\frac{t^2}{3} \right)$ 는 구간 $[0, 2x^2]$ 에서 연속이므로

적분의 평균값 정리를 이용하면 다음이 성립한다.

$$\frac{\int_0^{2x^2} \sin \left(\frac{t^2}{3} \right) dt}{2x^2 - 0} = \sin \left(\frac{c^2}{3} \right), \quad 0 < c < 2x^2$$

이때, $x \rightarrow 0$ 이면 $c \rightarrow 0$ 이다.

$$\text{즉, } \lim_{x \rightarrow 0} \frac{\int_0^{2x^2} \sin \left(\frac{t^2}{3} \right) dt}{x^6} = \lim_{x \rightarrow 0} \frac{\int_0^{2x^2} \sin \left(\frac{t^2}{3} \right) dt}{\frac{1}{2} x^4 \cdot 2x^2}$$

$$= \lim_{x \rightarrow 0} \frac{\sin \left(\frac{c^2}{3} \right)}{\frac{1}{2} x^4}$$

그러나 여기서 $c^2 \sim Kx^4$ 같은 비율 정보가 없기 때문에 극한을 결정할 수 없다.

8. $f(x) = \int_1^{2x+3} \frac{1}{t^4 - 2t^2 + 3} dt$ 일 때,
 $g'(0) = \frac{1}{f'(-1)} = \frac{1}{\frac{1}{(2x+3)^4 - 2(2x+3)^2 + 3} \times 2} \bigg|_{x=-1} = 1$ 이다.

9. $f(x) = (x^2 + 1)^2 - \int_0^x t^2 f'(t) dt$ 에서 양변을 미분하면

$$\Rightarrow f'(x) = 2(x^2 + 1)2x - x^2 f'(x)$$

$$\Rightarrow (1 + x^2)f'(x) = 4x(1 + x^2)$$

$$\Rightarrow f'(x) = 4x$$

$$\Rightarrow f(x) = 2x^2 + c \text{ 이고 } f(0) = 1 \text{ 이므로 } c = 1 \text{ 이다.}$$

$$\text{즉, } f(x) = 2x^2 + 1 \text{ 이므로 } f(2) = 9 \text{ 이다.}$$

10. $F(x) = \int_0^x f(t)(x-t)dt = x \int_0^x f(t)dt - \int_0^x tf(t)dt$ (양변을 미분)

$$\Rightarrow F'(x) = \int_0^x f(t)dt + xf(x) - x f(x) = \int_0^x f(t)dt$$

$$\Rightarrow F''(x) = f(x)$$

$$f > 0 \text{ 이므로 } F'(x) = \int_0^x f(t)dt > 0 \text{ 이고 } F''(x) = f(x) > 0 \text{ 이다.}$$

즉, $F(x)$ 는 증가함수이며 아래로 볼록인 함수이다.

11. (㉠) $f(a-x) = f(x)$ 일 때,

$$\int_0^a f(x)dx = \int_0^{\frac{a}{2}} f(x)dx + \int_{\frac{a}{2}}^a f(x)dx$$

$$= \int_0^{\frac{a}{2}} f(x)dx + \int_{\frac{a}{2}}^a f(a-x)dx$$

$$= \int_0^{\frac{a}{2}} f(x)dx - \int_{\frac{a}{2}}^0 f(t)dt \quad (\because a-x=t)$$

$$= \int_0^{\frac{a}{2}} f(x)dx + \int_0^{\frac{a}{2}} f(x)dx$$

$$= 2 \int_0^{\frac{a}{2}} f(x)dx \quad (\text{참})$$

[참고]

$f(a-x) = f(x)$ 를 만족하는 함수는 $x = \frac{a}{2}$ 축 대칭이므로

$$\int_0^a f(x)dx = 2 \int_0^{\frac{a}{2}} f(x)dx \text{가 항상 성립한다.}$$

$$(㉡) \int_{-a}^a f(x)dx = \int_{-a}^0 f(x)dx + \int_0^a f(x)dx$$

$$= - \int_a^0 f(-t)dt + \int_0^a f(x)dx \quad (\because x=-t)$$

$$= \int_0^a f(-x)dx + \int_0^a f(x)dx$$

$$= \int_0^a f(-x) + f(x)dx \quad (\text{참})$$

$$(㉢) \int_0^\pi xf(\sin x)dx = - \int_\pi^0 (\pi-t)f(\sin t)dt \quad (\because x=\pi-t)$$

$$= \int_0^\pi (\pi-x)f(\sin x)dx$$

$$= \int_0^\pi \pi f(\sin x)dx - \int_0^\pi xf(\sin x)dx$$

$$\text{이므로 } 2 \int_0^\pi xf(\sin x)dx = \int_0^\pi \pi f(\sin x)dx \text{ 이다.}$$

$$\text{즉, } \int_0^\pi xf(\sin x)dx = \frac{\pi}{2} \int_0^\pi f(\sin x)dx \quad (\text{참})$$

$$(㉣) \int_0^{\frac{\pi}{2}} f(\sin x)dx = - \int_{\frac{\pi}{2}}^0 f(\cos t)dt \quad (\because x=\frac{\pi}{2}-t)$$

$$= \int_0^{\frac{\pi}{2}} f(\cos x)dx \quad (\text{참})$$

12. $\int_0^{\frac{\pi}{2}} \frac{\sin^n x + \cos^n x}{\sin^n x + \cos^n x} dx = \frac{\pi}{2}$ 이므로

$$\int_0^{\frac{\pi}{2}} \frac{\sin^n x}{\sin^n x + \cos^n x} dx + \int_0^{\frac{\pi}{2}} \frac{\cos^n x}{\sin^n x + \cos^n x} dx = \frac{\pi}{2} \text{ 이다.}$$

$$\text{이때, } \int_0^{\frac{\pi}{2}} \frac{\sin^n x}{\sin^n x + \cos^n x} dx$$

$$= - \int_{\frac{\pi}{2}}^0 \frac{\cos^n t}{\cos^n t + \sin^n t} dt \quad (\because x = \frac{\pi}{2} - t)$$

$$= \int_0^{\frac{\pi}{2}} \frac{\cos^n x}{\cos^n x + \sin^n x} dx \text{ 을 알 수 있다.}$$

$$\text{즉, } \int_0^{\frac{\pi}{2}} \frac{\sin^n x}{\sin^n x + \cos^n x} dx = \int_0^{\frac{\pi}{2}} \frac{\cos^n x}{\sin^n x + \cos^n x} dx = \frac{\pi}{4}$$

13. $\int_0^{\frac{\pi}{2}} \frac{\tan^n x + \cot^n x}{\tan^n x + \cot^n x} dx = \frac{\pi}{2}$ 이므로

$$\int_0^{\frac{\pi}{2}} \frac{\tan^n x}{\tan^n x + \cot^n x} dx + \int_0^{\frac{\pi}{2}} \frac{\cot^n x}{\tan^n x + \cot^n x} dx = \frac{\pi}{2} \text{ 이다.}$$

$$\text{이때, } \int_0^{\frac{\pi}{2}} \frac{\cot^n x}{\tan^n x + \cot^n x} dx$$

$$= - \int_{\frac{\pi}{2}}^0 \frac{\cot^n t}{\cot^n t + \tan^n t} dt \quad (\because x = \frac{\pi}{2} - t)$$

$$= \int_0^{\frac{\pi}{2}} \frac{\cot^n x}{\cot^n x + \tan^n x} dx \text{ 을 알 수 있다.}$$

$$\text{즉, } \int_0^{\frac{\pi}{2}} \frac{\tan^n x}{\tan^n x + \cot^n x} dx = \int_0^{\frac{\pi}{2}} \frac{\cot^n x}{\tan^n x + \cot^n x} dx = \frac{\pi}{4}$$

14. $\int_a^b [f(x)]^2 dx = 1$

$$\Rightarrow [xf^2(x)]_a^b - 2 \int_a^b xf(x)f'(x)dx = 1 \quad (\because \text{부분적분})$$

$$\Rightarrow bf^2(b) - af^2(a) - 2 \int_a^b xf(x)f'(x)dx = 1 \quad (\because \text{부분적분})$$

$$\Rightarrow -2 \int_a^b xf(x)f'(x)dx = 1$$

$$\text{즉, } \int_a^b xf(x)f'(x)dx = -\frac{1}{2}$$

15. $g(7) = 3, g(1) = 0 \Leftrightarrow f(3) = 7, f(0) = 1$ 이므로

$$\int_0^3 f(x)dx + \int_{f(0)}^{f(3)} g(x)dx = \int_0^3 f(x)dx + \int_1^7 g(x)dx$$

$$= 3 \times 7 - 0 \times 1 = 21 \text{ 이다.}$$

$$\text{즉, } \int_0^3 f(x)dx + 13 = 21 \text{ 이므로 } \int_0^3 f(x)dx = 8 \text{ 이다.}$$

16. $\int_1^2 f(x)dx + \int_{f(1)}^{f(2)} g(x)dx = \int_1^2 f(x)dx + \int_2^5 g(x)dx$

$$= 2 \times 5 - 1 \times 2 = 8 \text{ 이다.}$$

$$\text{이때, } \int_1^2 \frac{1}{3}x^3 + \frac{2}{3}x + 1dx + \int_2^5 g(x)dx = 8 \text{ 이므로}$$

$$\int_2^5 g(x)dx = 8 - \int_1^2 \frac{1}{3}x^3 + \frac{2}{3}x + 1dx$$

$$= 8 - \left[\frac{1}{12}x^4 + \frac{1}{3}x^2 + x \right]_1^2$$

$$= 8 - \left(\frac{4}{3} + \frac{4}{3} + 2 - \frac{1}{12} - \frac{1}{3} - 1 \right) = \frac{19}{4} \text{ 이다.}$$

[다른 풀이]

$$g(x) = y \text{ 라하면 } f(y) = x \text{ 이고,}$$

$$f(1) = 2, f(2) = 5 \Rightarrow g(2) = 1, g(5) = 2 \text{ 이므로}$$

$$\int_2^5 g(x)dx = \int_1^2 yf'(y)dy$$

$$= \int_1^2 y \left(y^2 + \frac{2}{3} \right) dy = \left[\frac{1}{4} y^4 + \frac{1}{3} y^2 \right]_1^2 = \frac{19}{4} \text{이다.}$$

$$\begin{aligned} 17. & \int_0^3 \frac{1}{f'(f^{-1}(x))(1+(f^{-1}(x))^2)} dx \\ &= \int_0^1 \frac{1}{f'(t)(1+t^2)} f'(t) dt \quad (\because f^{-1}(x) = t) \\ &= \int_0^1 \frac{1}{1+t^2} dt = [\tan^{-1} t]_0^1 = \frac{\pi}{4} \end{aligned}$$

[다른 풀이]

$f(f^{-1}(x)) = x$ 에서 양변을 미분하면

$$f'(f^{-1}(x)) \cdot (f^{-1})'(x) = 1 \Rightarrow (f^{-1})'(x) = \frac{1}{f'(f^{-1}(x))} \text{이다.}$$

$$\begin{aligned} \text{즉, } & \int_0^3 \frac{1}{f'(f^{-1}(x))(1+(f^{-1}(x))^2)} dx \\ &= \int_0^3 \frac{(f^{-1})'(x)}{(1+(f^{-1}(x))^2)} dx = [\tan^{-1}(f^{-1}(x))]_0^3 \\ &= \tan^{-1} 1 - \tan^{-1} 0 = \frac{\pi}{4} \text{이다.} \end{aligned}$$

$$18. \int_1^2 t f(t) dt \text{는 상수값이므로}$$

$$\int_1^2 t f(t) dt = C \text{라 하면 } f(x) = \ln x + C \text{이고}$$

$$\begin{aligned} C &= \int_1^2 t(\ln t + C) dt = \int_1^2 t \ln t + C t dt \\ &= \left[\frac{1}{2} t^2 \ln t - \frac{1}{2} \int t dt + \frac{1}{2} t^2 C \right]_1^2 \\ &= \left[\frac{1}{2} t^2 \ln t - \frac{1}{4} t^2 + \frac{1}{2} t^2 C \right]_1^2 = 2 \ln 2 - 1 + 2C + \frac{1}{4} - \frac{1}{2} C \end{aligned}$$

$$\Leftrightarrow -\frac{1}{2} C = 2 \ln 2 - \frac{3}{4}$$

$$\Leftrightarrow C = -4 \ln 2 + \frac{3}{2} \text{이다.}$$

$$\text{따라서 } f(x) = \ln x - 4 \ln 2 + \frac{3}{2} \text{이므로 } f(2) = -3 \ln 2 + \frac{3}{2} \text{이다.}$$

장황수학

1. 2번	2. 3번	3. 3번	4. 1번	5. 4번
6. 3번	7. 2번	8. $\frac{1}{\sqrt{2}}$	9. 3번	10. 1번
11. 1번	12. e^{-1}	13. 2번	14. 5번	15. 2번
16. 3번	17. 1번	18. 2번	19. 4번	

1. (1) 심프슨 공식

$$\frac{\Delta x}{3} \left(f(0) + 4f\left(\frac{\pi}{4}\right) + 2f\left(\frac{\pi}{2}\right) + 4f\left(\frac{3\pi}{4}\right) + f(\pi) \right)$$

$$= \frac{\pi}{12} (2\sqrt{2} + 2 + 2\sqrt{2}) = \frac{\pi}{6} (2\sqrt{2} + 1)$$

(2) 사다리꼴 공식

$$\frac{\Delta x}{2} \left(f(0) + 2f\left(\frac{\pi}{4}\right) + 2f\left(\frac{\pi}{2}\right) + 2f\left(\frac{3\pi}{4}\right) + f(\pi) \right)$$

$$= \frac{\pi}{8} (\sqrt{2} + 2 + \sqrt{2}) = \frac{\pi}{4} (\sqrt{2} + 1)$$

$$\Rightarrow 6A - 4B = \pi(2\sqrt{2} + 1) - \pi(\sqrt{2} + 1) = \sqrt{2}\pi$$

2. 가. $\left| \int_a^b f(x) dx \right| \leq \int_a^b |f(x)| dx$ (참)

나. $F(-x) = \int_x^{-x} f(t) dt = - \int_{-x}^x f(t) dt = -F(x)$ (참)

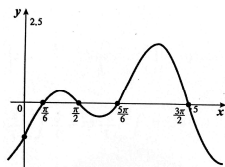
다. [반례]

$f(x) = 3x - 1$ 이라 하면, $\int_0^1 3x - 1 dx \leq 0$ 이지만

$x > \frac{1}{3}$ 일 때, $f(x) > 0$ 이다. (거짓)

라. 만약 모든 x 에서 $f(x) \leq 0$ 이면 적분값도 ≤ 0 이므로 모순이다.
따라서 어떤 점에선 반드시 $f(x) > 0$ 이다. (참)

3. $f(x) = \sin 2x - \cos x = 2\sin x \cos x - \cos x = \cos x (2\sin x - 1)$



$\int_a^b f(x) dx$ 가 최대가 되도록 하는 실수 a, b 에 대한

$b - a$ 의 최댓값은 $a = \frac{\pi}{6}, b = \frac{3\pi}{2}$ 이므로 $b - a$ 의 최댓값은 $\frac{4}{3}\pi$

4. $\int_1^n \frac{1}{x^p} dx \leq \frac{1}{1^p} + \frac{1}{2^p} + \frac{1}{3^p} + \dots + \frac{1}{n^p} \leq 1 + \int_1^n \frac{1}{x^p} dx$ 이므로

$$\lim_{n \rightarrow \infty} \int_1^n \frac{1}{x^p} dx \leq \lim_{n \rightarrow \infty} \left(\frac{1}{1^p} + \frac{1}{2^p} + \frac{1}{3^p} + \dots + \frac{1}{n^p} \right)$$

$$\leq \lim_{n \rightarrow \infty} \left(1 + \int_1^n \frac{1}{x^p} dx \right)$$

$\therefore \int_1^\infty \frac{1}{x^p} dx \leq S_p \leq 1 + \int_1^\infty \frac{1}{x^p} dx$

$\therefore \frac{1}{p-1} \leq S_p \leq \frac{p}{p-1}$

5. $f(x) = \int_0^1 |t^2 - x^2| dt$

$= \int_0^1 (x^2 - t^2) dt \quad (\because 0 \leq t \leq 1, |x| \geq 1)$

$= \left[x^2 t - \frac{1}{3} t^3 \right]_0^1 = x^2 - \frac{1}{3}$

이때, $f(x) = x^2 - \frac{1}{3}, x \leq -1, x \geq 1$ 에서의

최솟값은 $f(-1) = f(1) = \frac{2}{3}$ 이다.

6. $g(e) + g\left(\frac{1}{e}\right) = \int_1^e \frac{\ln t}{1+t} dt + \int_1^{\frac{1}{e}} \frac{\ln t}{1+t} dt$

이때, $\int_1^{\frac{1}{e}} \frac{\ln t}{1+t} dt$ 에서 $t = \frac{1}{x}$ 로 치환하면

$\int_1^e \frac{\ln \frac{1}{x}}{1 + \frac{1}{x}} dx = \int_1^e \frac{-\ln x}{x(x+1)} dx$ 이다.

즉, $g(e) + g\left(\frac{1}{e}\right) = \int_1^e \frac{\ln t}{1+t} dt + \int_1^{\frac{1}{e}} \frac{\ln t}{1+t} dt$

$$= \int_1^e \frac{\ln t}{1+t} dt + \int_1^e \frac{\ln t}{t(1+t)} dt$$

$$= \int_1^e \frac{(1+t)\ln t}{t(1+t)} dt = \int_1^e \frac{\ln t}{t} dt = \left[\frac{1}{2} (\ln t)^2 \right]_1^e = \frac{1}{2}$$

7. $\theta = \pi - x$ 라 하면

$\int_0^\pi \theta f(\sin \theta) d\theta = \int_\pi^0 (\pi - x) f(\sin(\pi - x)) (-dx)$

$= \int_0^\pi (\pi - x) f(\sin x) dx$

$= \pi \int_0^\pi f(\sin x) dx - \int_0^\pi x f(\sin x) dx$

$= \pi \int_0^\pi f(\sin \theta) d\theta - \int_0^\pi \theta f(\sin \theta) d\theta$

$\therefore \int_0^\pi \theta f(\sin \theta) d\theta = \frac{\pi}{2} \int_0^\pi f(\sin \theta) d\theta$

8. $\int_0^\pi \frac{x \sin x}{1 + \sin^2 x} dx = \frac{\pi}{2} \int_0^\pi \frac{\sin t}{1 + \sin^2 t} dt$

$= \frac{\pi}{2} \int_0^\pi \frac{\sin t}{2 - \cos^2 t} dt = \frac{\pi}{2} \int_{-1}^1 \frac{1}{2 - \theta^2} d\theta \quad (\because \cos t = \theta)$

$= -\pi \int_0^1 \frac{1}{\theta^2 - 2} d\theta = \frac{-\pi}{2\sqrt{2}} \left[\ln \left| \frac{\theta - \sqrt{2}}{\theta + \sqrt{2}} \right| \right]_0^1$

$= \frac{\pi}{2\sqrt{2}} \ln \left| \frac{1 + \sqrt{2}}{1 - \sqrt{2}} \right| = \frac{\pi}{2\sqrt{2}} \ln \left| \frac{1 + \sqrt{2}}{1 - \sqrt{2}} \right|$

$= \frac{\pi}{2\sqrt{2}} \ln(\sqrt{2} + 1)^2 = \frac{\pi}{\sqrt{2}} \ln(\sqrt{2} + 1)$

즉, 주어진 문제의 정답은 : $\frac{1}{\sqrt{2}}$

9. $I_{2018} = \int_0^\pi \frac{\sin 2018x}{x} dx \quad (u = \pi - x \text{로 치환})$

$= \int_\pi^0 \frac{\sin 2018(\pi - u)}{\sin(\pi - u)} du$

$= \int_0^\pi \frac{\sin(2018\pi - 2018u)}{\sin u} du = \int_0^\pi \frac{-\sin 2018u}{\sin u} du$ 이므로

$\int_0^\pi \frac{\sin 2018x}{\sin x} dx = - \int_0^\pi \frac{\sin 2018x}{\sin x} dx$

$\Leftrightarrow 2 \int_0^\pi \frac{\sin 2018x}{\sin x} dx = 0 \Leftrightarrow \int_0^\pi \frac{\sin 2018x}{\sin x} dx = 0$

$I_{2017} = \int_0^\pi \frac{\sin 2017x}{\sin x} dx = \int_0^\pi \frac{\sin(2016x + x)}{\sin x} dx$

$= \int_0^\pi \frac{\sin 2016x \cos x + \cos 2016x \sin x}{\sin x} dx$

$$\begin{aligned}
&= \int_0^\pi \frac{\sin 2016x \cos x}{\sin x} dx + \int_0^\pi \cos 2016x dx \\
&= \int_0^\pi \frac{\frac{1}{2} \{\sin(2017x) + \sin(2015x)\}}{\sin x} dx + \frac{1}{2016} [\sin 2016x]_0^\pi \\
&= \frac{1}{2} \int_0^\pi \frac{\sin 2017x}{\sin x} dx + \frac{1}{2} \int_0^\pi \frac{\sin 2015x}{\sin x} dx \quad \text{이므로} \\
\int_0^\pi \frac{\sin 2017x}{\sin x} dx &= \frac{1}{2} \int_0^\pi \frac{\sin 2017x}{\sin x} dx + \frac{1}{2} \int_0^\pi \frac{\sin 2015x}{\sin x} dx \\
\Leftrightarrow \frac{1}{2} \int_0^\pi \frac{\sin 2017x}{\sin x} dx &= \frac{1}{2} \int_0^\pi \frac{\sin 2015x}{\sin x} dx \\
\Leftrightarrow \int_0^\pi \frac{\sin 2017x}{\sin x} dx &= \int_0^\pi \frac{\sin 2015x}{\sin x} dx \quad \text{이다.}
\end{aligned}$$

따라서

$$\int_0^\pi \frac{\sin 2017x}{\sin x} dx = \int_0^\pi \frac{\sin 2015x}{\sin x} dx = \dots = \int_0^\pi \frac{\sin x}{\sin x} dx = \pi$$

이다. 그러므로 $|I_{2018} - I_{2017}| = \pi$ 이다.

$$\begin{aligned}
10. \lim_{n \rightarrow \infty} \sum_{k=1}^n f_n(k) &= \lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{k^2 - 3kn + n^2}{n^3} \\
&= \lim_{n \rightarrow \infty} \sum_{k=1}^n \left[\left(\frac{k}{n} \right)^2 - \frac{3k}{n} + 1 \right] \frac{1}{n} \\
&= \int_0^1 x^2 - 3x + 1 dx = -\frac{1}{6}
\end{aligned}$$

$$\begin{aligned}
11. \lim_{n \rightarrow \infty} \sum_{k=n}^{3n} \frac{1}{n+k} &= \lim_{n \rightarrow \infty} \sum_{k=n}^{3n} \frac{1}{\frac{k}{n} + 1} \frac{1}{n} \\
&= \int_1^3 \frac{1}{1+x} dx = [\ln(1+x)]_1^3 = \ln 2
\end{aligned}$$

$$\begin{aligned}
12. \lim_{n \rightarrow \infty} \left(\frac{n!}{n^n} \right)^{\frac{1}{n}} &= \lim_{n \rightarrow \infty} e^{\frac{\ln(n!)}{n}} \\
&= \lim_{n \rightarrow \infty} e^{\frac{1}{n} \ln(n!)} = \lim_{n \rightarrow \infty} e^{\frac{1}{n} \ln(1 \cdot 2 \cdot 3 \cdots n)} \\
&= \lim_{n \rightarrow \infty} e^{\frac{1}{n} (\ln 1 + \ln 2 + \ln 3 + \dots + \ln n)} \\
&= \lim_{n \rightarrow \infty} e^{\frac{1}{n} \sum_{k=1}^n \ln k} = e^{\int_0^1 \ln x dx} = e^{-1}
\end{aligned}$$

[다른 풀이]

스털링 근사를 활용하면

$$\begin{aligned}
\lim_{n \rightarrow \infty} \left(\frac{n!}{n^n} \right)^{\frac{1}{n}} &\sim \lim_{n \rightarrow \infty} \left(\frac{\sqrt{2\pi n} n^n e^{-n}}{n^n} \right)^{\frac{1}{n}} = \lim_{n \rightarrow \infty} \left(\frac{(\sqrt{2\pi n})^{\frac{1}{n}}}{e} \right) = \frac{1}{e} \quad \text{이다.} \\
(\because (\sqrt{2\pi n})^{\frac{1}{n}} &= e^{\frac{\ln \sqrt{2\pi n}}{n}} = e^0 = 1)
\end{aligned}$$

$$\begin{aligned}
13. g_n'(x) &= \sum_{k=1}^n \left[f' \left(\frac{k}{n} x \right) \frac{k}{n} \frac{x^2}{n} + f \left(\frac{k}{n} x \right) \frac{2x}{n} \right] \\
\Rightarrow \lim_{n \rightarrow \infty} g_n'(x) &= \lim_{n \rightarrow \infty} \sum_{k=1}^n \left[f' \left(\frac{k}{n} \right) \frac{k}{n} \frac{1}{n} + f \left(\frac{k}{n} \right) \frac{2}{n} \right] \\
&= \int_0^1 [x f'(x) + 2f(x)] dx \\
&= \int_0^1 (x e^x + 2e^x) dx = \int_0^1 (x+2)e^x dx \\
&= [(x+2)e^x]_0^1 - \int_0^1 e^x dx = 2e - 1
\end{aligned}$$

14. 정적분의 평균값 정리를 만족시키는 점 ξ 를 찾으려 한다.

$$f(\xi) = \frac{1}{\frac{3}{2}\pi - 0} \int_0^{\frac{3}{2}\pi} \sin 2x dx = \frac{2}{3\pi} = \sin 2\xi$$

따라서 $\sin 2a = \frac{2}{3\pi}$, $\left(0 < a < \frac{\pi}{4}\right)$ 라고 할 때,

$$\xi = a, \xi = \frac{\pi}{2} - a, \xi = \pi + a, \xi = \frac{3}{2}\pi - a$$

이므로 모든 ξ 값의 합은 $\frac{\pi}{2} + \frac{5}{2}\pi = 3\pi$ 이다.

$$15. \frac{\int_x^{f(x)} \frac{3+2t^2}{1+t^2} dt}{f(x)-x} = \frac{3+2c^2}{1+c^2}, \quad x < c < f(x)$$

$$\Leftrightarrow f(x) - x = \frac{1+c^2}{3+2c^2}$$

$$\therefore \lim_{x \rightarrow \infty} (f(x) - x) = \lim_{c \rightarrow \infty} \frac{1+c^2}{3+2c^2} = \frac{1}{2}$$

$$16. 넓이 = f(t) = \int_t^{2t} \sin x dx \text{의 최댓값}$$

$$\Rightarrow f'(t) = 2 \sin 2t - \sin t = 4 \sin t \cos t - \sin t = \sin t (4 \cos t - 1)$$

$$\Rightarrow t = \cos^{-1} \frac{1}{4} \quad \left(\because 0 < t < \frac{\pi}{2} \right)$$

$$\begin{aligned}
\Rightarrow f\left(\cos^{-1} \frac{1}{4}\right) &= \int_{\cos^{-1} \frac{1}{4}}^{2 \cos^{-1} \frac{1}{4}} \sin x dx \\
&= -\left[\cos 2\left(\cos^{-1} \frac{1}{4}\right) - \cos\left(\cos^{-1} \frac{1}{4}\right) \right] = \frac{9}{8}
\end{aligned}$$

$$17. A(t) = \frac{1}{2} \cosh t \sinh t - \int_1^{\cosh t} \sqrt{x^2 - 1} dx$$

$$A'(t) = \frac{1}{2} (\sinh^2 t + \cosh^2 t) - \sqrt{\cosh^2 t - 1} \sinh t$$

$$= \frac{1}{2} (\sinh^2 t + \cosh^2 t) - \sinh^2 t$$

$$= \frac{1}{2} (\cosh^2 t - \sinh^2 t) = \frac{1}{2}$$

$$\begin{aligned}
18. \lim_{x \rightarrow 1} \frac{\int_1^x (x-t)f(t)dt}{(x-1)^2} &= \lim_{x \rightarrow 1} \frac{x \int_1^x f(t)dt - \int_1^x t f(t)dt}{(x-1)^2} \\
&= \lim_{x \rightarrow 1} \frac{\int_1^x f(t)dt}{2(x-1)} = \lim_{x \rightarrow 1} \frac{f(x)}{2} = \frac{f(1)}{2} \quad (\because \text{로피탈 정리}) \\
&= \frac{1+1+1^2+\dots+1^{99}}{2} = 50
\end{aligned}$$

19. (i) $x > 0$ 일 때,

$$\int_{-x}^x (x^2 - t^2 + |t|)f(t)dt = x|x|$$

$$\Leftrightarrow \int_{-x}^x (x^2 - t^2 + |t|)f(t)dt = x^2$$

$$\Rightarrow x^2 \int_{-x}^x f(t)dt + \int_{-x}^x (-t^2 + |t|)f(t)dt = x^2$$

양변을 x 로 미분하면

$$\Rightarrow 2x \int_{-x}^x f(t)dt + x^2 \{f(x) + f(-x)\}$$

$$+ (-x^2 + |x|)f(x) + (-x^2 + |-x|)f(-x) = 2x$$

$$\Rightarrow 2x \int_{-x}^x f(t)dt + x \{f(x) + f(-x)\} = 2x \quad (\because x > 0)$$

$$\Rightarrow 2 \int_{-x}^x f(t)dt + \{f(x) + f(-x)\} = 2$$

양변을 x 로 미분하면

$$2\{f(x) + f(-x)\} + \{f'(x) - f'(-x)\} = 0$$

$$\Rightarrow \frac{f'(x) - f'(-x)}{f(x) + f(-x)} = -2$$

양변을 적분하면

$$\ln |f(x) + f(-x)| = -2x + C$$

$$\Rightarrow f(0) = 1 \text{ 이므로 } C = \ln 2$$

$$\Rightarrow \ln|f(x) + f(-x)| = -2x + \ln 2$$

$$\text{즉, } f(x) + f(-x) = e^{-2x + \ln 2} \text{이므로 } f(1) + f(-1) = 2e^{-2}$$

(ii) $x < 0$ 일 때도 같은 방법으로 구할 수 있다.

[다른 풀이]

$x > 0$ 일 때,

$$\int_{-x}^x (x^2 - t^2 + |t|)f(t)dt = x|x|$$

$$\Leftrightarrow \int_{-x}^x (x^2 - t^2 + |t|)f(t)dt = x^2 \text{에서}$$

라이프니츠 정리를 활용하면

$$|x|f(x) + |-x|f(-x) + \int_{-x}^x 2xf(t)dt = 2x$$

$$\Leftrightarrow x(f(x) + f(-x)) + \int_{-x}^x 2xf(t)dt = 2x \text{이다.}$$

$$\begin{aligned} \text{이때, } \int_{-x}^x 2xf(t)dt &= \int_{-x}^0 2xf(t)dt + \int_0^x 2xf(t)dt \\ &= \int_0^x 2xf(-\alpha)d\alpha + \int_0^x 2xf(t)dt \quad (\because t = -\alpha) \\ &\Rightarrow \int_0^x 2xf(-t)dt + \int_0^x 2xf(t)dt \\ &= 2x \int_0^x (f(t) + f(-t))dt \text{이므로} \end{aligned}$$

$$x(f(x) + f(-x)) + \int_{-x}^x 2xf(t)dt = 2x$$

$$\Leftrightarrow x(f(x) + f(-x)) + 2x \int_0^x (f(t) + f(-t))dt = 2x$$

$$\Leftrightarrow (f(x) + f(-x)) + 2 \int_0^x (f(t) + f(-t))dt = 2 \quad (\because x \neq 0)$$

이제 $f(x) + f(-x) = g(x)$ 라 하면

$$\Leftrightarrow g(x) + 2 \int_0^x g(t)dt = 2$$

$$\Leftrightarrow g'(x) + 2g(x) = 0 \quad (\because \text{양변 미분})$$

$$\Leftrightarrow \frac{g'(x)}{g(x)} = -2$$

$$\ln|g(x)| = -2x + C \text{에서}$$

$$f(0) + f(-0) = g(0) = 2 \text{이므로 } C = \ln 2 \text{이다.}$$

$$\text{즉, } g(x) = e^{-2x + \ln 2} \text{이므로}$$

$$\therefore f(1) + f(-1) = g(1) = 2e^{-2}$$

